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SYSTEM 21

TECHNICAL INFORMATION

Version 3.5.2



FRONTISPIECE

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This information relates to Application Version 3.5.2

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Glossary of Terms

Glossary

Acrobat Printing Problems

JBA issue Help files in .pdf format, for example wm.pdf, for use with Acrobat Viewer. You can print pages directly from Acrobat Viewer, however, there may be a problem where Arial Monospaced font has been used to show excerpts of program code and the printer is not set up correctly to receive this font.

In order for all fonts to print correctly on your printer you must make sure of the following two things:

- That you have the correct print resolution set up on your PC. You must have 300 dots per inch.
- That you only print from your hard drive or a network drive.

If you do not know how to do this then please take the following steps. You need only do this once:

1. Close Acrobat viewer.
2. Load the .pdf file onto your hard drive on the PC, or to a network drive.



Note: If the .pdf file has been sent to you via Lotus Notes then please detach to the hard drive3. Select Start\Settings\Printers to display the Printers Window.

4. Select your default printer and click with the right mouse button. This displays the pop up menu.
5. Select Properties from the pop up menu.
6. Select the panel with the Resolution field (usually the Graphic panel).
7. Change the Resolution to 300 dots per inch.
8. Click Apply and then OK.
9. Open Acrobat Viewer and resume printing from your .pdf file as normal.

Activities

There are two types of Activities:

- Assembled Activities. These are stored in .ACT files.
- Elemental Activities. These are stored in the design-time database.

App21Res

Changed using the Preferences dialogue. As with Sys21Res, but applies to called PC Applications as opposed to Host sessions/tasks. Refer also to Limitations Of The PC.

Assembled Activities

User defined activities which are broken down into Elemental Activities, Business Methods and Subassemblies.

Business Methods

These are the lowest level software components and cannot be broken down any further. You find these contained within Assembled Activities and Subassemblies.

Business Methods can be Constructables, Consumables or both of these.

Business Processes

These are at the top level of the business function.

Within a Business Process it is possible to have Elemental Activities, Assembled Activities and other Business Processes. Business Processes can be nested to several levels.

However, there are restrictions on the way in which you construct Business Processes depending on whether you use Constructor as a design tool for Enterprise Modelling or a configuration tool for Work Management.

If the Business Process is to be uploaded to Work Management these rules apply:

- Assembled activities cannot be present.
- Nesting can only be to one level down.
- At the top level, a nested Business Process must be linked to an end node.
- A User Selected Path cannot link into a Parallel Path.

There may be other restrictions in place, but the validation will highlight these.

ChangeUIFont

You set this in Jbasmg.ini, the Configuration File for System 21 Explorer. You can set this to 1 or 0 or omit any reference to ChangeUIFont.

ChangeUIFont=1 causes Explorer and Session Manager to use the same font as that used by the Emulator.

ChangeUIFont=0 causes Explorer and Session Manager to use the design time fonts and this is usually an ANSI font. ANSI fonts are suitable in the UK, USA and Western Europe but are not suitable in Central Europe, Japan or China.

Omitting any reference has the same result as ChangeUIFont=0.

CHP Error -1001 CHPHostNotFound

The host specified by the name parameter was not found.

CHP Error -1002 CHPTryAgain

The local server does not receive a response from an authorised server. Try again later.

CHP Error -1003 CHPNoRecovery

Unrecoverable error. Check that TCP/IP is installed, configured properly and started.

CHP Error -1004 CHPConnRefused

Connection refused. Check that the correct port number for the CAS service is specified in the TCP/IP services file and that the Job Queue server is alive on the UNIX server.

CHP Error -1005 CHPNetUnreached

The network is unreachable.

CHP Error -1006 CHPTimedOut

The connection timed out.

CHP Error -1007 CHPNetDown

The network is down.

CHP Error -1008 CHPPFNoSupport

Protocol family not supported (in this case TCP/IP).

CHP Error -1009 CHPAFNoSupport

Address family not supported by the protocol family.

CHP Error -1010 CHPAddrInUse

The IP address of the PC is already in use.

CHP Error -1011 CHPAddrNotAvail

Cannot assign requested address.

CHP Error -1012 CHPNetReset

Network dropped connection on reset.

CHP Error -1013 CHPConnAborted

Software caused cpmnectopm abort.

CHP Error -1014 CHPConnReset

Connection reset by peer.

CHP Error -1015 CHPNoBufs

No buffer space available.

CHP Error -1016 CHPIsConn

The socket is already connected.

CHP Error -1017 CHPNotConn

The socket is not connected.

CHP Error -1018 CHPHostDown

The host you are trying to connect to is down.

CHP Error -1019 CHPHostUnreach

No route to the host.

CHP Error -1020 CHPNotFound

The program was not found on the host. Check that the library specified was correct or that it is contained in the library list for your session.

CHP Error -1021 CHPIntr

Interrupted system call.

CHP Error -1022 CHPBadF

Bad file number.

CHP Error -1023 CHPAcces

Permission denied.

CHP Error -1024 CHPFault

Bad Address.

CHP Error -1025 CHPInval

Invalid argument.

CHP Error -1026 CHPMfile

Too many open files.

CHP Error -1027 Local Logon

There is no local logon.

CHP Error -1028 Server/Host

Unable to retrieve a either a server name or a host name from the configuration files using the given client application. Please check that your configuration is correct.

CHP Error -1029 UserId

No server user id was configured for the current local logon. Please check that your configuration is correct.

CHP Error -1099 CHPUnknown

Unknown error.

CHP Error -2000 CHPNullProgram

The program name is blank.

CHP Error -2001 CHPPassword

Invalid password.

CHP Error -2002 CHPService

It was not possible to resolve the jbachp service. Check that there is an entry for this in your services file. Also make sure that you are looking at the right services file CHP Error - it should be in the directory specified by the ETC environment variable.

CHP Error -2003 CHPPipeOpen

Pipe open failed on the server.

CHP Error -2004 CHPWriteOnSocket

Write error on socket.

CHP Error -2005 CHPInvalidType

Invalid Type parameter to CallUnixProgram (this currently defaults to 1 and cannot be modified in JOT).

CHP Error -2006 CHPInvalidLength

Invalid length (>1024) read from the socket by the server.

CHP Error -2007 CHPFilePermissions

The user does not have the permissions necessary to find the file.

CHP Error -2008 CHPPathComponent

A component of the path is not a directory.

CHP Error -2009 CHPNameTooLong

The file name is too long.

CHP Error -2010 CHPBrokenPipe

Broken pipe. There is something wrong with the connection to the server CHP Error - reconnect and try to call the program again.

Client

Usually a PC, an intelligent terminal which sends data to, and receives data from, the Server.

Client Workspace

JBA System 21 Client Workspace is 16bit software which provides a single point of access to JBA System 21 applications and to those applications that run under Microsoft Windows.

It launches tasks through user configurable Soft Menus. These Soft Menus are pull down options from the menu bar.

There is an advanced facility which enables the System Administrator to reformat the application windows (see Client Workspace Designer).

Client Workspace Designer

This is an integral part of JBA System 21 Client Workspace. When you install Client Workspace you also install Designer. Designer enables the System Administrator to tailor the System 21 window display for the Client Workspace user.

Clipping

Clipping is the term used to describe the way in which lines can be hidden from the screen display.

Cloud

This is a colloquial term for the data exchange area.

Colours

Changed by the Colour panel. Please note that if the JBAS21P.INI file is deleted, the colours will be reset to BLACK. This may be corrected by taking the Colour dialogue, selecting default, and exiting System 21 again. The correct colours will then be saved.

Component

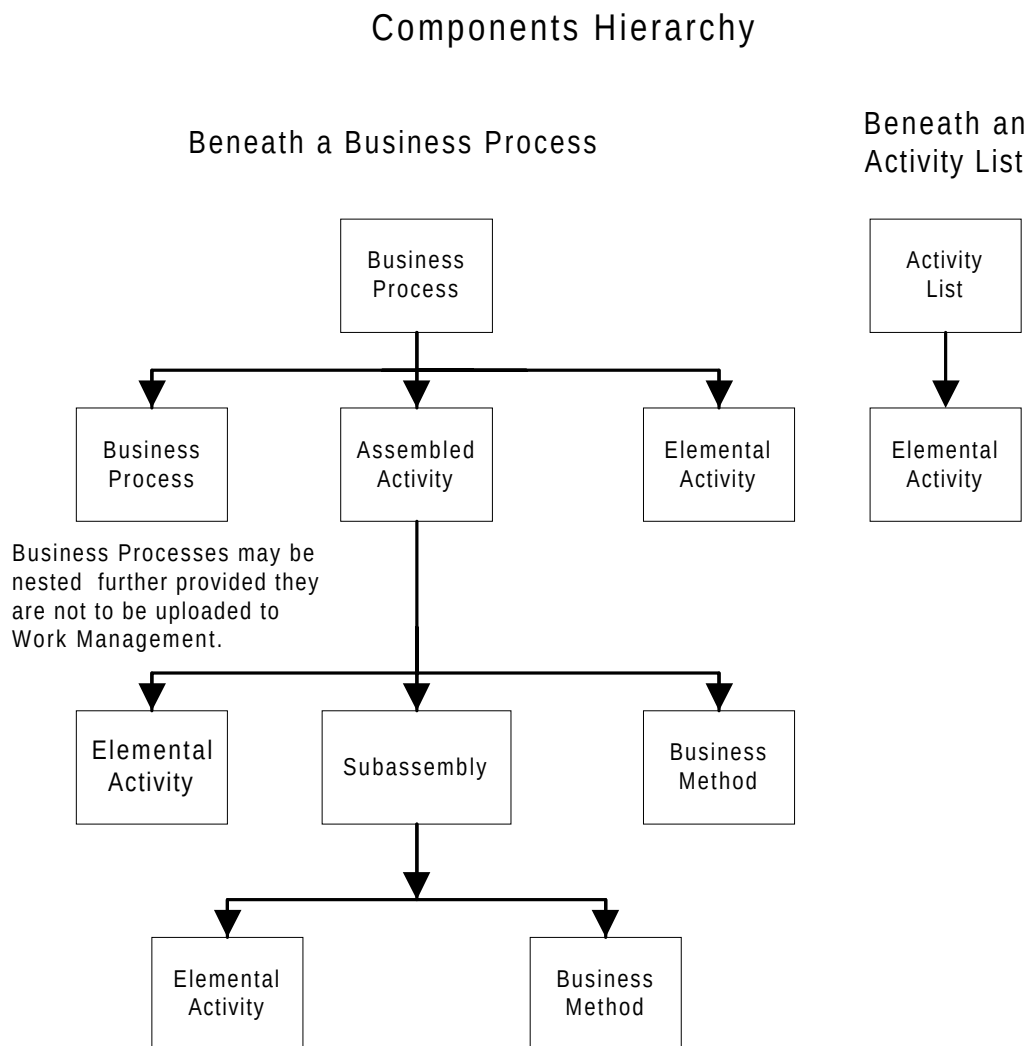
A Component is a part of the design of a business system. It can be an Assembled Activity, an Elemental Activity, a Business Method, a Constructable or a Consumable.

Within Constructor components are grouped hierarchically:

- The top level in the hierarchy consists of the Business Processes.
- The middle level consists of the Assembled Activities, Elemental Activities, Activity Lists and Subassemblies.

- The bottom level consists of the Business Methods.

Components Hierarchy Diagram



Connection Protocol

There are three options:

AS400 - You connect to an AS/400 and you use a router (for example, IBM Client Access or NetSoft)

AS400 TCPIP - You connect to an AS/400 using TCP/IP

UNIX or NT - You connect to Retrofit

Connections

These are lines which link the components together to form a continuous process.

Constructable

A constructable is a component which has to be called and executed. It can be an Elemental Activity or a Business Method. It can contain Consumables.

Constructor

Constructor is a framework generator tool that you use to construct your business system using a hierarchy of components. It integrates fully with Work Management and System 21 Explorer. Using a Windows environment you model your system design on a Workspace, incorporating System 21 application components. You decide the components and then link them together hierarchically to form a logical business structure.

Components are grouped hierarchically:

- The top level in the hierarchy consists of the Business Processes.
- The middle level consists of the Assembled Activities, Elemental Activities, Activity Lists and Subassemblies.
- The bottom level consists of the business methods.

A design should ideally be started at the bottom level so that the top level finally incorporates all the lower levels. Please refer to the Components Hierarchy diagram. You can use Constructor in one of three ways:

- As a design tool for the construction of Assembled Activities.
- As a design tool for Enterprise Modelling.
- As a graphical design and configuration tool for Work Management. You configure the design of a Business Process and upload it to the Work Management Engine. Work Management then controls the flow of work through Explorer by initiating activities and managing Action Lists.

System 21 enables a customer to integrate standard System 21 components in conjunction with their own bespoke components, based on Microsoft ActiveX technologies. At each stage in the design process you can validate the design so that you can be certain that it is viable.

You can preview and execute Business Methods, Activities, Activity Lists and Subassemblies from within Constructor. Although you cannot directly execute a Business Process in Constructor you can perform a walkthrough to verify the flow of control between the configured components. You can launch the underlying System 21 application components individually.

Consumable

A consumable is an executable Business Method which is used to contain programming function which can easily be changed. For example a selection of tax codes and calculations would be stored as consumables so that different processing could be applied when a user changes a tax code.

Countries

Not normally changed. However, in some countries, the CodePage varies according to the Host Type. For the Czech Republic, the AS/400 Page is 870 (as shipped) , but should be changed to 912 for a UNIX connection. The local Agent should know what these settings should be and can advise if a Manual change is required.

Client Workspace: Virtual Workstation for UNIX (JBAVT) Defaults. The UNIX implementation of Client Workspace uses the UNIX JBA Virtual Workstation DLL (Jbavt.dll) . The Jbas21p.ini file may require customising for AS/400 hosts depending on the router used.

Explorer: Virtual Workstation for UNIX (JBAVT) Defaults. The UNIX implementation of Explorer uses the UNIX JBA Virtual Workstation DLL (Jvtm1d.dll) . The jbasmg.ini file may require customising for AS/400 hosts depending on the router used.

CSS Error -01 Communications Router Not Installed

The Client/Server support routines can find no trace of the communications router in memory. Associated CSS Constant is DbNoRouterInstalled.

CSS Error -02 Communications Router Not Started

Either the communications router has not been started, or a previous error on the local PC or the host has terminated the communications link. Associated CSS Constant is DbNoConnection.

CSS Error -03 No More File Handles Available

An attempt has been made to concurrently open more files than the Client/Server routines allow. The current maximum number of files is set at 30. Associated CSS Constant is DbNoMoreFiles.

CSS Error -04 CPIO program not found

The CPIO program specified in the file open request could not be found on the host system.

CSS Error -05 Unable to Load Required DLLs

One of the DLLs required for the application could not be loaded. Associated CSS Constant is DbDLLUnavailable.

CSS Error -06 No More Connection Handles Available

The maximum number of connections allowed for the application has been reached. Associated CSS Constant is DbNoMoreConnections.

CSS Error -07 Invalid Scan of Table

The description for the file specified in one of the file open functions could not be retrieved. The format of the data (from the host) is invalid or an error code from the host was not recognised. Associated CSS Constant is DbScanTableError.

CSS Error -08 Invalid Host Type

The host type specified for the connection attempt is not supported. Associated CSS Constant is DbInvalidHost.

CSS Error -09 No FFD File Found

No Local File Field Description (FFD) File Found.

When accessing DB/2 or Database manager, the field description information for the table or view being opened could not be located. Associated CSS Constant is DbNoFieldDefinitions.

CSS Error -10 Invalid File

Attempt to perform a record operation or file close with an invalid file handle. Associated CSS Constant is DbInvalidFile.

CSS Error -11 File Not Found

The file specified in an open request could not be found on the host. Associated CSS Constant is DbUnknownFile.

CSS Error -12 Invalid Program Name

The program you are attempting to run could not be found or its name is invalid for the host. Associated CSS Constant is DbInvalidProgram.

CSS Error -13 Invalid Operation For File

The operation specified is invalid for the file type. For example, a keyed operation on a nonkeyed file. Associated CSS Constant is DbInvalidOperation.

CSS Error -14 Invalid Key Depth For File

The key depth specified for the operation is either less than one or is greater than the number of key fields in the file. Associated CSS Constant is DbInvalidKeyDepth.

CSS Error -15 File Already Open

The file specified in the open request is already open. Associated CSS Constant is DbAlreadyOpen.

CSS Error -16 Too Many Files Open

The maximum number of open files allowed for this application has already been reached. Associated CSS Constant is DbTooManyFiles.

CSS Error -17 Invalid Numerics

Invalid Number of Records For File MultiCSS Error -read.

CSS Error -18 Call Host Program Failed

The call to the specified program has failed. This could be due either to the program not being found in the specified library, or to the program failing due to a logical error. Associated CSS Constant is DbCallFailed.

CSS Error -19 Execute Host Command Failed

The call to the specified command has failed. This could be due either to the command not being found in the specified library, or to the command failing due to a logical error. Associated CSS Constant is DbCommandFailed.

CSS Error -20 File Override Failed

File override failed

CSS Error -21 Open Query File Failed

The open query file function failed because the query parameters passed to the function call contained either an invalid parameter or option. Associated CSS Constant is DbQueryFailed.

CSS Error -22 CPIO Program Failed

An attempt to open a file with a CPIO program (and library) failed. CPIO programs are called when reads and writes and so on are performed on the file. Thus the PC can perform normal file access whilst the AS/400 actually emulates record access with an (AS/400) program.

CSS Error -23 Open File Failed

File open failed (on a valid connection). The file failed to open. Possible reasons include the file not being found in the library list or the library specified, or the file itself was of an unsupported type. Associated CSS Constant is DbOpenFailed.

CSS Error -24 File Not Open

Attempt to perform a record access or file close on a handle which could be but is not valid (e.g. a handle for a file which has been closed) Associated CSS Constant is DbFileNotOpen.

CSS Error -25 No Record(s) Found

No more records left in multi read buffer or record not found. Either the record specified in a keyed get to the file was not found or the last record in a multiCSS Error -read operation has already been fetched. Associated CSS Constant is DbNoRecordFound.

CSS Error -26 No More Records

No More Records for Current MultiCSS Error -read Operation. Fetch record attempted without multi read.

CSS Error -27 Write Record Failed

The attempt to add a record to the file has failed. This could be due to a keyed file having a unique key attribute and an attempt being made to add a record with a key that already exists. Associated CSS Constant is DbWriteFailed.

CSS Error -28 No Records With Equal Key

When setting the file pointer for the file, no record exists with a key equal to that specified in the call, but records with keys greater than or less than the specified key exist. Associated CSS Constant is DbNoRecordEqual.

CSS Error -29 Update Record Failed

The attempt to update a record in the file has failed. This could be due to a keyed file having the unique key attribute and an attempt being made to update a record with a new key value that already exists. Associated CSS Constant is DbUpdateFailed.

CSS Error -30 Delete Record Failed

The attempt to delete a record from the file failed. This could be due to a keyed delete being requested, and the specified record not existing.

CSS Error -31 Locked Record

An attempt has been made to update, delete or fetch a record with update lock which is already locked by a different process. Associated CSS Constant is DbRecordLocked.

CSS Error -32 Unauthorised Access

You are not authorised to the JBA Open ToolCASE product on the host computer. Associated CSS Constant is DbUnauthorisedAccess.

CSS Error -33 Unauthorised Action or Operation

You have attempted to access a function, or execute a program or procedure to which you are not authorised. Associated CSS Constant is DbNotUnauthorised.

CSS Error -34 No Commitment Control on Host

A Commit or Rollback function has been requested for a session or host that does not have commitment control installed. Associated CSS Constant is DbNoCommitmentControl.

CSS Error -35 Commit Failed

The requested commit operation has failed. Refer to messages on the host for details of the error. Associated CSS Constant is DbCommitFailed.

CSS Error -36 Rollback Failed

The requested rollback operation has failed. Refer to messages on the host for details of the error. Associated CSS Constant is DbRollbackFailed.

CSS Error -37 File Close Failed

The file failed to close. This could be due to the file not being open. Associated CSS Constant is DbRollbackFailed.

CSS Error -38 Create Local File Failed

The creation of the local file failed. Either the file name or path specified is invalid, or there is not enough disk space available. Associated CSS Constant is DbCreateFailed.

CSS Error -39 Read Failed

This error code is returned from DB2/2 functions when an error occurs executing an SQL statement. One possible cause is deadlock. Note, this error code may be returned on a write or a delete also.

CSS Error -40 Invalid Parameter

One of the parameters specified in the function call is invalid. Refer to the documentation for the function in error for details of the correct parameters. Associated CSS Constant is DbInvalidParameter.

CSS Error -50 Level Check Error

The level of the file that the program was created over is not the same as the level of the file that is currently being opened.

CSS Error -60 No Local Application List Present

The locally held applications file cannot be found. File Applicat.Ini could not be found in the JBAFILES directory. In order to determine which host is to be connected to and which user profile to use to access that host, local configuration files are used. One of these files is the list of local applications. Each of the records in this file defines a local application. This file will not exist until you have defined at least one application. If you would like to run any of the local configuration utilities now, select the appropriate item from below&colon.CSS Error -

CSS Error -61 Unknown Local Application

The local application specified in the Connect To Application function call does not exist in the local applications file. If you would like to run any of the local configuration utilities now, select the appropriate item from below&colon.CSS Error - Associated CSS Constant is DbUnknownLocalApplication.

CSS Error -70 Required Local Logon Not Present

Under OS2, this means that there is no local logon. Under Windows, this means that the environment variable CSSUSER has not been set. In order to determine which host is to be connected to and which user profile to use to access that host, local configuration files are used. One of these files is the list of users for a particular host. Each of the records in this file defines a local user ID and the relevant host user ID and password to use when making the connection. The local user ID specified for the host configured in the Connect To Application function call is not logged on locally. Associated CSS Constant is DbNoLocalLogon.

CSS Error -71 No Host User Entry

No user IDs have been defined for the host configured for the application specified in the Connect To Application function call. In order to determine which host is to be connected to and which user profile to use to access that host, local configuration files are used. One of these files is the list of users for a particular host. Each of the records in this file defines a local user ID and the relevant host user ID and password to use when making the connection. This file will not exist until you have defined at least one host. If you would like to run any of the local configuration utilities now, select the appropriate item from below. CSS Error - Associated CSS Constant is DbNoHostUserEntry.

CSS Error -72 No Host User List

File HostUser.Dat could not be found. Ensure that file exists in the directory pointed to by environment variable JbaFiles.

CSS Error -73 No Host User Entry

No entry for the current user (local user for OS/2, CSSUSER for Windows) could be found.

CSS Error -74 No User Host List

File UserHost.Dat could not be found. Ensure that file exists in the directory pointed to by environment variable JbaFiles.

CSS Error -80 Function/Operation Not Supported

This function is not supported on this host. All file access to a local file now return this error code. Associated CSS Constant is DbUnsupportedFunction.

CSS Error -81 Memory Request Failed

A memory allocation failed. If this error occurs under Windows, it is probably a shortage of conventional memory. You could try running MemMaker (a Microsoft DOS utility) to optimise conventional memory.

CSS Error -90 The Connection Attempt Failed

Details for host were found in CSS configuration but connection to host could not be made using these details. When the connection to the host is made, certain information has to be passed to the host in order for the link to be made. If any of this information is missing or incorrect, then an error occurs. e.g. Partner Local User name (PLU) could be invalid. Associated CSS Constant is DbConnectionFailed.

CSS Error -97 Locking Error

Error getting per instance memory from the lock layer. Associated CSS Constant is DbMemoryError.

CSS Error -98 Synonym Error

Use DBDebugOn or set SynDbg=2 for more details. Associated CSS Constant is DbSynonymError.

CSS Error -99 Communications Error

Please set CspDbg=2 and view trace file for more details Associated CSS Constant is DbCommunicationsError.

CWSINITIAL

This is a setting in the Configuration Files which determines the startup program. There are two possible settings:

- Program=CWSINITIAL. This is the default and in most cases you can accept this.
- Program=*NONE. For this you would have to edit the Configuration File and change CWSINITIAL to *NONE. See below for further explanation.

For System 21 entry the initial program is set to the startup program. If the Sign On Panel does not have a field to enter an initial program, each System 21 User must be changed to have their initial program changed to CWSINITIAL (change the User Profile INLPGM parameter) . To prevent System 21 attempting to add the program into the non-existent field on the Sign On panel, the above variable must be set to Program=*NONE

Please note that if the User has an initial program of AMINITIAL, or a customised sign-on program which uses AMINITIAL to access System 21, then the Host User Profile need not be changed.

For the maximum performance benefit, it is recommended to use CWSINITIAL. However, if set, the UserID can only be used for the Client Workspace or Explorer (it cannot be used to sign on to a dumb terminal or other emulation session).

This must be done for any user set up as CLASS *USER or with LIMITCAPABILITIES set to *YES or *PARTIAL.

Data Exchange Area

A storage area on the computer which is used for passing parameters (variables) from one piece of JBA program code to another at runtime, regardless of deployment. This is also known, colloquially, as the cloud.

Data Source

A ODBC definition which shows the location of the data brought back by the ODBC. Please refer to the ODBC Glossary Definition.

DDE

Dynamic Data Exchange

DEA

Data Exchange Area

Default Option

Client Workspace And Explorer

Set via the Countries option on the System menu within Client Workspace, and defaulting to UKB. This will dictate which codepage and character set to use.

Designer

JBA have two Designer products:

- Client Workspace Designer
- Explorer Designer.

DesignFiles

Within the Client Workspace Designer, it is necessary to indicate whether SRC or SRU SRfiles are being maintained. 1 refers to the SRC, and 2, the SRU files. The default is taken from this .INI variable, with an installed default of 2. The setting last used is saved to this variable when Client Workspace is closed down.

DLL

Dynamically Linked Libraries.

Dynamic Data Exchange.

This is a interface which enables one program to talk to another. For example Microsoft Word and System 21 Explorer both have DDE interfaces which means that Word can be called from System 21 Explorer and documents accessed.

To do this you use the sticky fields facility where you are allocated a field in System 21 to be accessible to the third party application.

EHNAPPC Router to AS/400 Table

JVWDBG Value	Description	
1	Traces out unexpected results	
2	Traces interaction with the LUE	
4	Non-categorised trace	
8	WorkStation function control logic trace	
16	Passthrough interface trace	
32	Trace surrounding EHNAPPC function calls	
64	The actual bytes sent/recv'd from the host	
128	5250 Data stream translation trace	
256	Internal data storage trace	
512	Character translation trace	
1024	Monitor queries for change from host	

Elemental Activities

JBA activities on the server which cannot be broken down any further

Emulator Trace Table

LUEDBG Value	Description	
1	Traces out unexpected results	
2	Show calls to and from application	
4	Basic sequence of drawing	
8	Memory Management	
16	Communications with host	
32	Monitor queries for change from host	

64	Intercept messages (eg key presses)	
128	Alignment and drawing of fields	
256	Recognition of Pop Ups and Buttons	
512	Simple formatting of screens	
1024	Don't hide the entry point panel	
2048	Measure performance (tiny amount)	
4096	Turns on trace for fonts	
8192	Graphs out internal lists	

EWM

Enterprise Work Management.

Explorer

Explorer is 32 bit software and provides a single point of access to JBA System 21 applications and to those applications that run under Microsoft Windows.

Applications are launched from a series of role menus, designed to reflect the activities performed by the people within the organisation. For example there could be one role menu for all the Sales Order Clerks, another one for the Inventory Clerks and so on.

There are advanced facilities, designed for a System Administrator, which can be switched on and off :

- Role Menu Designer enables you to design role menus to tailor your system to meet your business needs so that individual users can be screened from sensitive applications.
- System Configuration enables you to configure the system for the users.
- Explorer Designer enables you to reformat the application windows.
- User/Role Maintenance enables you maintain a list of users in the database and a list of authorised role menus for each user.
- Document Status Review enables you to view the current status of any Business Object on your database. You may also launch a pending Document Type.
- Document Viewer enables you to maintain a database of documents which can then be linked to System 21 Application Windows, Elemental Activities or third-party programs.
- Email Server Agent runs unattended and forwards Email messages generated by the Work Management engine.

Explorer Designer

This is an integral part of System 21 Explorer. When you install Explorer you also install Designer. Designer enables the System Administrator to tailor the System 21 window display for the Explorer user.

Font

There are three fonts:

- Arial Monospaced
- Arial Monospaced CE
- Arial Monospaced Zero-Dot

In Client Workspace each font may have either a bold or normal style making a total of six combinations.

In Explorer you can use a bold override setting, using the Preferences Screen Window. This updates the BoldFont= setting in Jbasmg.ini from 0 to 1.

Fonts are saved as .ttf files in the installed software directory with a subdirectory \FONTS.

FrameSize

Initial startup frame size as a portion of the screen. The default is 0.9, covering 90% of the screen. For lower screen resolutions, it may be preferable to set this to 1.0 for full screen.

Gid

global id

Goal Posted

This expression is used to describe windows which do not display properly, often with just three sides of black, on a grey border - hence the name. However, more commonly it is used to describe any window with an incorrect display.

HostName

Client Workspace only

It defaults to Host, and Client Workspace will overwrite it with the LAST HOST accessed normally. This is picked up from the Router during connection, and confirmed by the User at logon. There is no need to set this field prior to starting Client Workspace.

InfoBase Help

This is comprehensive online help text provided by JBA to accompany their software products. When you press the Help button on any JBA application window you are presented with help for that window. However, from there you can go on to navigate the help for the entire suite of JBA applications.

IniEnvironment

Applicable to Client Workspace only.

Changed using the Preferences dialogue. Set to 0 (default) , the User's environment is taken from the Host. Whenever Client Workspace is exited, the current Host and Environment are saved at the foot of the JBAS21P.INI file. Setting IniEnvironment to 1 will force Client Workspace to use the environment specified in the .INI file (if it has an environment specified for the connected Host). Failure to find one will revert to the Host User settings.

Input Field

This is a field into which you type data.

JbaDTab.Dat

The single byte file which by default is loaded for AS400 TCPIP connections when the host is a DBCS host.

JbaGTab.Dat

The file which by default is loaded for AS400 TCPIP connections. Simply copying one of the other files over this file will cause all AS400 TCPIP (which have not been separately configured) connections to use this Translation File.

Language

Language or Country code, changed using Country dialogue. See Character Translation.

LanguageFile

Language Translation file - shipped as JBAS21.UK, but can be changed to any translated equivalent.



Warning: This file is critical. If amended incorrectly it may cause Client Workspace or Explorer failures. Please refer to your JBA consultant.

Lid

local id

Limitations Of The PC

The System 21 interface is as resource and memory intensive as most common big name applications. Running alongside these applications will increase the risk of the PC reaching its processing limits. Opening many sessions on the PC will reduce this still further.

It is recommended that each PC be tested for its limitations, and then set so that in normal use, this is not reached. With other desired tasks running, see how many sessions can be opened. If the maximum number of sessions was not achievable, then set the MaxSessions to the number of sessions obtained. If insufficient sessions can be opened for desired use, then other applications will have to be closed to allow this functionality to be within the limitations of the PC. Increasing the PC specification will raise the limitation, allowing more applications to be run, and more sessions to be opened. See: MaxSessions.

For Client Workspace, for a UNIX host, the main limitation is the number of DOS file handles. Client Workspace is restricted as the TCP/IP connections also take up file handles.

LoadBalance

Applicable to Client Workspace only.

If there is a Load Balancing program for UNIX, it can be enabled by this variable. The LoadBalanceProg and LoadBalancePath should indicate what program is to be run:

- Set to 0 (default) , there is no Load Balancing
- Set to 1 initiates Load Balancing on startup of Client Workspace
- Set to 2 also includes a display of load balancing results

Refer to UNIX Load Balance for help with running a Load Balance program.

LU2 Error -16 to -20

Please contact JBA Product Support for the latest information.

Please refer to Checklists for more information.

LU2 Error -15 LU2_DBCS_TRAN

Did not load a required DBCS translation table.

Please refer to Checklists and Character Translation for more information.

LU2 Error -14 LU2_ALREADY_OPEN

Attempt to open a session that is already open.

Please refer to Checklists for more information.

LU2 Error -13 LU2_INHIBITED

Cannot send because keyboard is inhibited.

Please refer to Checklists for more information.

LU2 Error -12 LU2_NO_SESSION

Session has not been started. No channel.

Please refer to Checklists for more information.

LU2 Error -11 LU2_NOT_STARTED

Comms have not been started. Usually caused when a required DLL was not loaded. For example: E32Appc.Dll for Win32, Ehnappc.dll for Win16 - for AS/400, WinSock.Dll for AS/400 TCP/IP or NT/Unix.

This could be a path problem or a configuration of a third party product.

Please refer to Checklists for more information.

LU2 Error -10 LU2_NO_HOST_INI

No default host found in the INI file

The details for the host you are trying to connect to are not present in the INI file. This error would only normally be produced when running a test harness. Ensure that the host name is present in the INI file. Try using an INI file from another PC.

Please refer to Checklists for more information.

LU2 Error -9 LU2_NO_ROUTER

Attempt to open session on a router not loaded

This is where there is an attempt to connect to a machine with a router which has not been loaded. It is an internal error. It could be caused by changing the settings for this host in the INI file in an inconsistent way.

Please refer to Checklists for more information.

LU2 Error -8 LU2_NULL_STRING

NULL string passed to SetConnectionMode

This is an internal error where no router name has been passed to an internal function.

Please refer to Checklists for more information.

LU2 Error -7 LU2_BAD_ROUTER

There is no router with specified name

This is an internal error where the router asked for was not loaded. This could be due to mixing new software, which supports a new router, with old software, which does not support a new router.

Please refer to Checklists for more information.

LU2 Error -6 LU2_BAD_CHARACTER

Wrong character used to identify router DLL

This is an internal error where the router name is not found even though it follows the normal naming convention. This could be due to mixing new software, which supports a new router, with old software, which does not support a new router.

Please refer to Checklists for more information.

LU2 Error -5 LU2_QUERY_FAILED

Failed calling Jvx's QueryCommsStatus

This is caused when a routine request to ask the router for an update fails. This could be because the network has failed or because something unexpected has been received over the network or because the query has been made at wrong time. Press Yes to continue. However, if the error continues you must end the session. The state of the network or third party router should be checked at this time to make sure that it is still working correctly.

This error could also be caused if the Job on the Host fails - check for logs on the AS/400 using DspJobLog and DspMsg QSysOpr.

For more information please refer to Routers and Checklists.

LU2 Error -4 LU2_NO_SYSTEM

Last system reached in list from Vw

This is an internal error. When using a third party router, Client Access or NetSoft, it is possible to get a list of machines which can be connected to. This return code indicates that there are no more system names to query.

Please refer to Checklists for more information.

LU2 Error -3 LU2_DEFAULT_ERROR

Failed to create default session.

The default session could not be opened. Ensure that the default session on the third party router, Client Access or NetSoft, is connected.

Please refer to Checklists and Character Translation for more information.

LU2 Error -2 LU2_OPEN_ERROR

Error opening session.

Failed to connect to the host. Ensure that the host is working correctly. For more information please refer to Connection Errors in the Troubleshooting Checklist.

LU2 Error -1 LU2_SYSTEM_ERROR

If this error is displayed when you try to connect to the host, for example, when starting a host task, then please refer to Connection Errors in the Troubleshooting Checklist.

LU2 Error 0 LU2_INFORMATIONAL

For your information only.

Please refer to Checklists for more information.

LU2 Error 1 LU2_SEND_FAILED

This warning means that JVx failed to send data.

This means that the send of data to the host failed and you can try again. For example this might happen if you pressed F8 and the F8 did not get to the host.

Please refer to Checklists for more information.

LU2 Error 2 LU2_DEFAULT_TRAN

Translation problem.

This warning means that JVX failed to find the ASCII to EBCDIC character translation table.

The character translation table could be for example: jbagtab.dat. This is non-fatal because in the event of the error an internal translation table is used instead

Please refer to Checklists for more information.

LU2 Error 3 LU2_OEM_AND_FILE

When connecting to UNIX or NT, there are three options:

- no character translation is done (default)
- ANSI/OEM conversion is done
- a file driven conversion is done.

This warning indicates that the INI file was configured to perform both second and third options simultaneously and this is not allowed.

Please refer to Checklists for more information.

LueInfo

For Internal Use only.

LueInfo=0 is the default setting in Jbas21p.ini for Client Workspace.

LueInfoCfg

Variable containing the pathway to the magic number file Lueinfo.cfg.

Lueinfo.cfg is essential to keep the correct screen appearance of sessions. You can have a copy of Lueinfo.cfg on every PC or you can locate a copy centrally, as for a network installation, and then point to it with the variable LueInfoCfg.



Note: Locating the file centrally may have a performance impact, depending on where in the network the file is kept.



Tip: Lueinfo.cfg is frequently updated by software upgrades. For a network installation, JBA recommend you to hold the file centrally, and change the variable manually, before commencing Client installations. The Client installation will copy the changed Jbas21p.ini file locally, and you can then remove separate copies of Lueinfo.cfg.

MaxSessions

Maximum sessions that can be opened before a limit check error is displayed preventing further sessions being opened.

Client Workspace

Uses a maximum of 8 sessions. Opening extra sessions is resource intensive, and can result in GPFs or other memory type issues too. This can be set from 1 to 8, but the recommended range is 2 to 8. If Resource/Memory limitations are noticeable on a particular PC, lowering this setting to 3 or 4 effectively tailors the product to the PC. Note: 8 is the maximum number allowed, setting this higher will not provide more sessions, and may have knock-on effects.

Explorer

Uses by default a maximum of 8 sessions but you can change this to any number up to 20 by amending the manual update area of the jbasmg.ini file.

Constructor

Limited to the restrictions of the emulator and you cannot set it yourself. JBA also advise you not to open more than one copy of Constructor on your desktop.



Note: Refer also to Limitations Of The PC.

Monospaced Font

A Monospaced Font is a character set which uses single character spacing for all characters, regardless of their relative size. For example - the letters A, I and W could be regarded as average, narrow and wide in size - a monospaced font would give each of them the same amount of room on the screen.

NOTRAN.DAT

A Translation File which converts each character to itself. It is supplied as a starting point to develop custom translation files for UNIX or NT.

NumCountries

For system use only.

NumHosts

Signifies the number of UNIX Hosts detailed in the .INI file. All potential hosts must be listed in this section of JBAS21P.INI. Refer the section on Hosts below for further information. The file is shipped with default information for a single host.

ODBC

Open Database Connectivity. This is Microsoft defined technology and refers to the provision of standard interfaces to any database. ODBC is a vehicle for translating SQL statements, sending requests to the database and bringing back the data. For example a PC application might want to access an AS/400 database that is managed by the AS/400 operating system. The PC would use ODBC to interface to the database.

Microsoft provide an ODBC Administrator where you can determine the driver settings for each database. See the Settings\Control Panel option on your Start menu. If you have written your SQL statements correctly you can direct a machine to any database by changing the ODBC settings.

OEMCONV

Original Equipment Manufacturer Conversion.

OEM is similar to DOS - a preWindows system.

The setting in the Configuration File that determines whether an extra translation of all input is required before passing it to the System 21 interface for display. The norm is to make a single translation which is always carried out. The second is set by this option, and therefore defaults to no.



Warning: You must set OEMCONV=1 if you want use Character Translation.

If OEMCONV is not configured correctly then some characters not in Codepage 850 (UK/US) will be displayed incorrectly (for example, the £ sign).

Orthogonal

Containing right angles.

Output Field

This is a field used for displaying text. An example of this might be a field name or a heading.

OverrideTitleText

For Internal Use only.

Pathways

(*Path)

Changed using the Preferences paths dialogue. This is where Client Workspace will search to find additional objects it needs whilst running. These objects can be stored away from the main directory, for ease of use, to enable easy replacement or deletion, or to swap in and out (eg. for translation purposes).

PIP

Product Improvement Package

Process Controls

These are used to mark the start and end of the Business Processes and Activities and to influence the pathways between components. They include Start and end conditions, User Selected Paths, Parallel Paths, Manual Activities and Action Agents.

Prompt Button

By default, for the Prompt Button, System 21 Explorer uses a bitmap showing a pair of green binoculars. However, you can override this with any bitmap of your choice by amending the manual update area of the jbasmg.ini file.

There is a bitmap provided for you, which you may want to use, showing a pair of grey binoculars. Please refer to c:\Program Files\JBASys21\binogray.bmp.

Proportional Font

A Proportional Font is a character set which uses variable character spacing for all characters, depending on their relative size. For example - the letters A, I and W could be regarded as average, narrow and wide in size - a proportional font would adjust the spacing on the screen, so that the I would have less space than the A, and the W would have more space.

Recogniser

Enables the use of Screen Recognition Files, if they are loaded, in Client Workspace and Explorer.

Enables elements of screen reformatting:

- A number of lines at the top and bottom of the AS/400 screen can be removed (clipped) providing a more PC-like display.
- The periods that follow field descriptions, and function key numbers (such as F12), can be removed.

These settings affect all the screens which are not Visual Components. This is different to the Designer facility where you set options only for the active window.

Client Workspace

Uses files in SRUPath or SRCPath.

Select the Preferences option on the System Menu to display the Preferences Window. Select the System Button on the Preferences Window to display the Systems Window.

Explorer

Uses an Access database table called Designer.

Select the Preferences option on the Options Menu to display the Preferences Window. Select the Screen Panel and select the System Wide Settings.

Role

See also Role Code and Role Menu.



Note: This is not available in Client Workspace, this is an Explorer feature.

A role in System 21 Explorer is a job title to which certain activities are assigned. For example, a typical role might be that of Sales Order Clerk, to which the activities of order entry and order acknowledgement might have been assigned. The users who have been given Sales Order

Clerk permissions will then be able to perform these activities from the Sales Order Clerk's role menu.

Role Code

See also Role and Role Menu.

If you want to make full use of the Work Management facility then you must assign role codes to role menus. Each role menu has a unique role code.

Your System Administrator may want to give you these permissions:

- To define a role code for an existing role menu. To do this use the Role Code Window.
- To create a role code at the same time as creating a new role menu. To do this use the User Role Maintenance facility.

Role Menu

See also Role Code and Role.

A Role Menu is a collection of activities available to a person assigned to a particular role within the organisation.

System 21 Explorer provides the System Administrator with a comprehensive facility to design role menus to tailor a system to meet business needs. This facility can be switched on and off so that individual users can be screened from sensitive applications.

For example: You might have an Order Entry Clerk role menu designed specifically for the order entry clerk. On this menu would be all the activities which the order entry clerk would be expected to undertake, such as Order Entry; Order Cancellation; Enter Credit Notes; Stock Allocation. The PC used by this clerk would be enabled only for that role menu and each of the activities could be put onto the desk top for easy access.

Role menus are stored in .rmu files.

S21User

It defaults to User, and like HostName, is set by Client Workspace from the last User logged on. There is no need to set this field prior to starting Client workspace.

Samples

This is a directory which contains sample programs, provided by JBA to help you to set up interfaces between Explorer and external programs. When you install Explorer you have the option to install the samples to a Samples directory.

SelectHost

When using Manual Logon, this will prompt the User to select the desired Host to use. This is set in the Preferences panel, and defaults to ON. If set off, only the default host may be accessed by Manual Logon.

SessionIDLetter

Client Workspace and Explorer have the facility to customise the AS/400 Session ID Name. Most emulator products create a session ID of Pcname/S/1. So if the PC is recognised as ST001 for example, an emulator session created by it would be called ST001S1. Additional sessions will be ST001S2 etc. Many emulator products will not connect to that host if the appropriate *S1 ID is being used by another emulator. With this facility, Client Workspace will connect and create a different ID name - shipped defaulting to J. The above PC will now create a session called ST001J1. In the unlikely event of another emulator using the same session Id (J) , this variable can be changed to any letter (including S if preferred) to circumvent the problem. With so many options, it should be possible to connect Client Workspace to any AS/400 regardless of what emulator is using it.

SessionSize

You can set this in Jbamsq.ini, the Configuration File for System 21 Explorer. It is in the [JBAS21DEFAULTS] section of the file. You use this to change the default session window size. The value must be in the range 0.3 to 1, where SessionSize=1 displays the session window maximised.

SessSize

Applicable to Client Workspace only.

Initial startup session frame size as a portion of the Workspace. The default is 0.9, covering 90% of the Workspace. Setting this to 1.0 effectively makes the session as large as the Workspace can make it.

Sign On

If the standard Sign On Panel has been modified or translated, System 21 needs to be aware of the changes in order to recognise the sign on screen. A configurable file is provided so that you may describe your modified Sign On panel to Client Workspace or Explorer. To do this edit Jbas21p.ini for Client Workspace or jbasmg.ini for Explorer under the section called JBALU2.

The variables contained in the JBA Emulator defaults section may be modified so that the FIRST output field on the panel replaces the text Sign On.

Other variables may also be changed in a similar manner:

(First output field on...) WS=Sign On(Sign On Panel) WPM=Display Program Messages(Display Program Messages Panel) WM=Display Messages(Display Messages Panel) PE=Sign-on Information>Password Expired Panel) PX=Change Password(Change Password Panel)

All text must be an EXACT match to that on the screen (case sensitive). If not, Unexpected Panel messages may appear.

Also note that in the case of translated panels (especially) , if this has not been applied correctly, any task started will prompt a Sign On panel, and not run the task.

SMU

An acronym for Soft Menu Utility.

Snap

Not used.

Soft Menu

A term used to describe a menu which can be altered by the user.

SQL Error -0001 Not Started

Failed to start a session with a remote SQL server.

SQL Error -0002 Communication Error

Communications error between host application and remote SQL server. Communications subsystems may not be started on the local client, remote host or both.

Alternatively the client may have failed authentication on the remote host (either at the operating system level, the database server level or both).

SQL Error -0003 Incompatible Release Levels

Guidelines SQL support software and proprietary SQL server software are at incompatible levels.

SQL Error -0020 Database Not Found

An attempt was made to connect to a database that cannot be found. This may be returned when a DRDA connection is attempted through DB2/400. A connection has been established with the remote AS/400 but the connection to the remote database has failed.

Check that the database is available through the remote AS/400 by running the DSPRDBDIRE (Display Relational Database Directory Entries) command. This command displays a list of available databases.

SQL Error -0021 Database Not Supported

An attempt was made to connect to an unsupported SQL server.

SQL Error -0022 Error

An error whose cause cannot be determined has occurred.

SQL Error -1000 Session Not Started

Failed to start a session with a remote SQL server. Check the SQL Alias definition and the SQL Database Server definition.

SQL Error -1001 Session Already Started

An attempt was made to connect to a database that already had an established connection

SQL Error -1002 Database Not Located

An attempt was made to connect to a database that cannot be found.

SQL Error -1003 Database Not Closed

The connection with the SQL server failed to close successfully.

SQL Error -1004 Invalid Logon Data Area

The logon data area for the connection could not be found or is corrupted.

SQL Error -1005 Invalid Cursor Data Area

The Cursor data area could not be found or is corrupted. The application is using a cursor identifier that has been previously successfully passed to SQLClose() or the cursor identifier was not returned by a call to SQLSelect(), SQLSelectParam() or SQLExecParam().

SQL Error -1006 Invalid Bind Data Area

The Host variable bind area could not be found or is corrupted.

SQL Error -1007 Invalid SQL Statement

The syntax of the SQL statement is incorrect, check the SQLCA for more information.

SQL Error -1008 Exec Failed

SQLExec failed because the SQL statement is invalid, check the SQLCA for more information.

SQL Error -1009 Cursor Not Opened

SQLSelect was called but a cursor was not returned, check the SQLCA for more information. This may indicate that the remote SQL server has reached its limit of open cursors allowed for a single application.

SQL Error -1010 Operation Failed

The SQL operation failed, check the SQLCA for more information.

SQL Error -1011 Cursor Read Only

An attempt has been made to update/delete with a read only cursor.

SQL Error -1012 Column Not Found

The column named was not found in the select list of the associated statement.

SQL Error -1013 Datatype Mismatch

The SQL column type and the host variable type are not compatible.

SQL Error -1014 Convert Failed

Conversion between a SQL datatype and a JOT datatype failed. This may be due to a general mismatch of data types (converting a character type to a numeric type for example) or a problem of scale and precision between the SQL type and the JOT type.

SQL Error -1015 Datatype Not Supported

The SQL datatype is not supported on the remote SQL server.

SQL Error -1016 Column Does Not Exist

The column named was not found in the set of columns associated with the cursor.

SQL Error -1017 Operation Not Valid Now

An SQL verb was called out of sequence, eg SQLUpdate() before SQLFetch().

SQL Error -1018 Prepare Failed

The SQL server failed to prepare the SQL statement.

SQL Error -1019 Non Select Statement

The SQLSelect() or SQLSelectParam() statement parameter was not a select statement.

SQL Error -1020 Statement Too Long

The SQL statement is too long. Increase the statement buffer size in the SQL Alias definition.

SQL Error -1021 Exec Error

A call to SQLExec() or SQLExecValues() failed. Check the syntax of the SQL statement or that any conditions in the statement are valid.

SQL Error -1022 Fetch Required

An attempt was made to retrieve column values without fetching a row. SQLFetch(), or SQLGetRow() must be called before individual select list values may be retrieved.

SQL Error -1023 Not Update Mode

An attempt was made to update or delete with a read only cursor.

SQL Error -1024 Fetch Failed

Cursor fetch failed.

SQL Error -1025 Invalid Cursor

The cursor based operation failed because the cursor does not exist.

SQL Error -1026 Open Cursor Failed

A cursor could not be opened against the database.

SQL Error -1027 Last Row

The last row for the cursor has been returned. All the rows in the active set of rows corresponding to a previously executed SQLSelect() or SQLSelectValues() have been retrieved by calls to SQLFetch(), SQLFetchValues() or SQLGetRow().

SQL Error -1028 Maximum Cursors Opened

The maximum allowed number of cursors for this application are in use.

SQL Error -1029 Communication Error

Communications error between host application and remote SQL server.

SQL Error -1030 Incompatible Release Levels

Guidelines SQL support software and proprietary SQL server software are at incompatible levels.

SQL Error -1031 Parameter Markers

An attempt was made to match a JOT application variable to a parameter markers and a datatype mismatch has occurred. A variable passed in a variable argument list may not be supported by the particular target database or the variable type may not be supported for the particular operation.

SQL Error -1032 Variable Not Bound

The application variable was not bound.

SQL Error -1033 Communication Error At Connection Time

Communications error between host application and remote SQL server.

SQL Error -1035 Maximum Connections

The maximum number of connection's allowed for a particular SQL server database are in use, this limit is imposed by the database itself.

SQL Error -1036 Maximum Cursors

The maximum number of cursors as allowed by the remote serve are in use.

SQL Error -1037 Invalid Parameter

One of the parameters passed to has a value that is out of range for the parameter.

SQL Error -1038 Non Select Cursor

The cursor belongs to a non SELECT SQL statement that has been prepared for execution. The cursor was passed to a function relevant to data retrieval and so the cursor is out of context for the requested operation.

SQL Error -1039 Too Many Variable Names In Parameter List

Too many variable names were passed to SQLSelectValues or SQLFetchValues.

SQL Error -1040 Too Few Variable Names In Parameter List

Too few variable names were passed to SQLSelectValues or SQLFetchValues.

SQL Error -2000 Maximum Connections

The maximum number of connections to remote SQL servers are in use.

SQL Error -2001 Invalid Logon Data Area

The Logon Data Area for a connection is corrupt.

SQL Error -2002 Logon Data Area Not Found

The Logon Data Area for a connection could not be found.

SQL Error -2100 Invalid Cursor Data Area

The Cursor Data Area associated with a cursor is corrupt.

SQL Error -2101 Maximum Cursors

The maximum number of cursors for a the specific SQL server connection are in use.

SQL Error -2102 Cursor Data Area Not Found

The Cursor Data Area associated with a cursor could not be found.

SQL Error -2103 Total Cursors Opened

The maximum number of cursors supported are in use.

SQL Error -2200 Bind Variable Not Found

The location of a variable bound to a SQL column could not be found.

SQL Error -2201 Maximum Bound

The maximum number of host variable have been bound.

SQL Error -2202 Already Bound

The column has already been bound to a host variable.

SQL Error -2300 DLLs Not Found

The DLLs required to support a specific SQL server could not be found.

Check that the Guidelines SQL support DLLs can be found by LIBPATH in CONFIG.SYS or are in the applications working directory.

Also the required communications software on the client may not be installed or similarly cannot be found by the LIBPATH variable in CONFIG.SYS.

SQL Error -2301 DLLs Not Loaded

The DLLs required to support a specific SQL server could not be loaded.

Check that the Guidelines SQL support DLLs can be found by LIBPATH in CONFIG.SYS or are in the applications working directory.

Also the required communications software on the client may not be installed or similarly cannot be found by the LIBPATH variable in CONFIG.SYS.

SQL Error -2302 DLLs Not Supported

The set of DLLs that were attempted to be loaded are not supported.

SQL Error -2303 DLLs Already Loaded

The set of DLLs to support a specific SQL server are already loaded.

SQL Error -2304 Maximum Logons

Maximum number of logons for a particular set of DLLs in use.

SQL Error -2305 Maximum DLL Sets Loaded

The maximum number of sets of DLLs are loaded.

SQL Error -2306 Failed To Load DLLs

In Windows, the operating system will tell you which DLL failed to load. In OS/2, call the JOT function SQLGetSQLCA and the name of the DLL which failed to load will be contained in the Msg part of the SQLCAType structure. Make sure that the directory which contains this DLL can be found on the LIBPATH in the config.sys.

SQL Error -2401 Undefined Local User

No local or global user was defined for the server which matches the currently logged on local user.

SQL Error -2402 Invalid Client Application

There is no client application of this name, or no server has been defined for it.

SQL Error -2403 No Local Logon

If you are using Windows, then make sure the CSSUSER environment variable is set.

SQL Error -2404 Database Name Not Defined

The database name edit field in the SQL section of the configuration utility is blank.

SQL Error -2405 Unable To Retrieve Database Info

The application was unable to retrieve information such as the DBMS mode or the isolation level. Switch on trace to find out which piece of information could not be found (see Troubleshooting section).

SQL Error -9000 Invalid Parameters

Parameter values for the function call are invalid.

Sticky Field

This is an input or output field where the value entered into it is saved to a storage area. Other activity windows or external applications can then read to or write from these sticky fields.

For example: Locating all details about an item may require the execution of several enquiries. By nominating the item reference as a sticky field in the first enquiry you could enable subsequent enquiries to be accessed without further rekeying of the item code.

You can transfer data to Microsoft Word by nominating, for example, the fields of an invoice as sticky fields and accessing them from Word using the DDE interface.

System 21 Interface

JBA provide interfaces to their System 21 applications and to those applications that run under Microsoft Windows:

- System 21 Client Workspace
- System 21 Explorer
- System 21 Constructor.

Separate online help text is provided with each of these interfaces.

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Sys21Res

Changed using the Preferences dialogue. If too much resource is used up in Windows, memory, allocation and resource issues have knock-on effects such as partial screen painting, performance degradation and GPFs. This can be limited by preventing the User from reaching critical thresholds. This will vary from PC to PC. 45 is a safe threshold for most PCs, and may be lowered to get greater functionality, but with greater risk, or raised for the reverse.

Also refer to Limitations Of The PC and App21Res.

System Configuration

This is installed alongside JBA System 21 Explorer as part of JBA's Administration Tools.

System Configuration enables the System Administrator to configure the options available to the person using Explorer or Constructor by enabling or disabling various facilities for them.

You can open System Configuration either from your Windows Start menu, by selecting Programs, JBA System 21, System Configuration or from the Preferences Window in System 21 Explorer or System 21 Constructor.

It runs expconfig.exe and updates the jbasmsg.ini file.

TCP/IP for AS/400 Table

JVNDBG Value	Description	
1	Traces out unexpected results	OK
2	Traces interaction with the LUE	?
4	Non-categorised trace	OK
8	WorkStation function control logic trace	OK
16	Passthrough interface trace	OK
32	Trace surrounding Winsock function calls	?
64	The actual bytes sent/recv'd from the host	?
128	5250 Data stream translation trace	?
256	Character translation trace	X
512	Monitor queries for change from host	X

TCP/IP Router for UNIX Table

JVTDBG Value	Description	
1	Traces out unexpected results	OK
2	Traces interaction with the LUE	OK
4	Non-categorised trace	OK
8	Traces the layer surrounding Winsock	OK
16	The actual bytes sent/recv'd from the host	??
32	Character translation/ Retrofit attributes	??
64	Monitor queries for change from host	OK
128	Measure performance (tiny amount of trace)	X

Technical Reference

An enquiry facility providing a complete cross-reference of server application programs, databases, activities and menus for JBA System 21 applications. It is designed for you to explore in detail the technical structure of the System 21 product.

It is primarily intended for Designers who want to perform an impact analysis. For example you might want to add a new field to the Item Master File and are aware that this could impact on every program which uses the file. You could use Technical Reference to find out the names of all the programs which use the Item Master File. This is a simple example but Technical Reference does enable you to navigate freely and make enquiries to numerous logical levels.

There are three standard environments: Generic, Drinks and Style. This software is installed as part of the Administration Tools installation, but you must purchase the databases separately.

TextOnly

Changed by the Preferences Panel. Shows the Function keys as Text only, or Text and Function key.

Tips And Techniques

Please assume that the Tips And Techniques section of the Help file refers to all three System 21 Interfaces, unless an individual interface is named specifically within the text.

Title

This refers only to the Title statement in the Explorer Configuration File Jbasmsg.ini.

In addition to specifying the text for the Explorer Title Bar you can also specify the font size to be used on the splash window and the Help Menu windows. You would use this only when installing a foreign language version of System 21 where the character set would cause unsightly text wrapping if the name was too long. For example, when installing the Japanese version of System 21 Explorer you would enter: Title=JBA System 21 Explorer,16.

Toolbar

Changed by the Preferences panel - this shows where the preferred screen location for the Business Process Toolbar should be.

TypeAhead

Two possible values:

- 1 - TypeAhead feature enabled (default).
- 0 - TypeAhead feature disabled.

UseCloud

You set this in the [JBAS21DEFAULTS] section of Jbamsgr.ini, the Configuration File for System 21 Explorer.

UseCloud=0 switches off the Data Exchange Area, or cloud. With the DEA switched off, the launching of assembled activities and activity list is not supported. If you have created external applications which directly access the DEA, either Sticky field or session data, you can convert them to use the external communication methods provided by Explorer.

Visual Component

This is a user interface program that adds a Windows-style compatible application interface to a character based application. Visual Components are developed as 32bit Visual Basic, Microsoft Windows components. They do not affect the server based business logic in any way and require no server side support. Visual Components work within the System 21 Explorer activity execution environment.

Work Management

The full title is Enterprise Work Management.

This is the term that describes JBA's implementation of activity task automation and control. The Work Management Engine is central to this processing and it is server-based.

Constructor is the Graphical Design/Configuration Tool for Work Management. You configure the design of a Business Process and upload it to the Work Management Engine in order to enable it for Work Management. Work Management then controls the flow of work through System 21 Explorer by initiating activities and managing Action Lists. However, there are restrictions on the way in which you construct Business Processes if they are to be uploaded because the uploading process involves rigorous validation checking.

System 21 Explorer is the interface which Work Management uses to initiate its enabled activities.

Work Management Engine

This is software which awaits incoming triggers which confirm the transactions that have been completed and may require the execution of a subsequent activity. There is a trigger for each completion state on a transaction, each one possibly generating a new pending transaction. The engine schedules new activities by putting them in a queue or adding them to a Action List. The engine supports parallel and serial execution of activities.

